What Contextual Holonomy Detects: the All-versus-Nothing Sector, Two Gap Certificates, and the Section Dependence of Wigner–Friend Holonomy

Manuel Flores Gordillo
Independent Researcher ORCID 0009-0006-8246-0343
manuel.flores@columbia.edu

July 2026

Abstract

The slogan "contextual holonomy encodes quantum contextuality" is attractive and, as usually stated, false. We give it a sharp, machine-verified form. On the context stack $[X/\mathrm{PU}(n)]$ of maximal abelian subalgebras the canonical degree-2 class has order exactly n; its cocycle holonomy along closed context loops detects *exactly* the all-versus-nothing (AvN) sector of contextuality for *closed-triple* Weyl context families at *even* local dimension (Theorem 1), a two-directional equivalence stress-tested on 2,100 random families at $d = 2, 4$ with zero mismatches. The evenness hypothesis is real and is stated as such (Remark 1); at odd d the achievable AvN value is 0 and the equivalence is not claimed. Both containments are strict: at $d = 3$ the class is nontrivial of order 3 while the achievable state-independent AvN value is 0 (Proposition 1, exhaustive certificate), and KCBS exhibits state-dependent contextuality where the class shadow vanishes (Proposition 2). Finally, the holonomy of the Wigner–friend loop generated by H and S splits by level: its projective (Pancharatnam) holonomy is $+1$, while its determinant-section holonomy is exactly $-i$, a τ-level invariant of the 2-adic tower (Observation 1). Interpretational claims must therefore specify loop and level. We state the residual equivalence problem precisely and give hardware-testable predictions. **Attribution.** The classification of state-independent values for qudit Weyl families — the even/odd dichotomy and the fixed value $d/2$ — on which the state-independent side of the equivalence rests is **due to Abramsky, Cercelescu and Constantin** (FSCD 2024) [4], and the general even-d *attainment* is the **Raussendorf–Okay–Zurel–Feldmann** construction [3]. This paper uses both and rediscovers neither; the recasting in Weyl-certificate language is in the companion [2]. **Status.** This is a preprint, not a peer-reviewed article. One claim of an earlier draft is withdrawn here (Retraction 1), and the hardware significances are shot-noise statistical only, with the caveat stated where they appear.

1 Setting and the repaired slogan

Let X be the space of maximal abelian $*$-subalgebras (MASAs) of $M_n(\mathbb{C})$, $n \geq 3$. The unitary group acts by conjugation; on $\mathrm{U}(n) \ltimes X$ the U(1)-lifting class is vacuous, and after Morita reduction the canonical class on $\mathrm{PU}(n) \ltimes X$ has order exactly n [1, Vacuity, Upper bound and Lower bound theorems] (`thmG_general.py`). Weyl context families and their commutator-carry invariant Q are as in [2]. Throughout, "verified" means: reproduced by the named script in the companion repository [7]; expected one-line outputs are pinned in `verification/INDEX.md`.

The repaired slogan is: *the class detects the AvN sector of contextuality, no more and no less.* The next three results make each clause exact.

2 The equivalence on the AvN sector

Fix $m \geq 1$ and a Weyl context family F whose contexts are commuting triples multiplying to a central element. Write the value-assignment system of F as $A\gamma \equiv s \pmod d$, where A is the context–observable incidence matrix and s collects the matrix-computed context phase exponents under the pinned Weyl convention.

Theorem 1 (Equivalence, AvN sector)**.** *Let d be* even. *Then F admits no noncontextual value assignment over $\mathbb{Z}_d$ if and only if some cycle of F carries odd anomaly self-pairing Q.*

Remark 1 (Scope, stated deliberately)*.* Two hypotheses are doing work here and neither is decoration.

Evenness. It is what makes the context phases automatically ω-powers (the parity lemma of [6]), and every source the proof rests on — [2, Thm. 8], the companion note's detection equivalence, and verification V44 — is stated at even d. At odd d the achievable state-independent AvN value is 0 (see Proposition 1 at $d = 3$), so the equivalence has no content there beyond that vanishing; we do not claim it, and the supporting evidence at odd d is sampling, not proof.

Closed triples. The odd-Q criterion is not asserted for arbitrary Weyl families: [2, Prop. 9] records that it can be incomplete in general. The families fixed above are closed-triple, which is the regime the certificates cover, and `shadow_gap.py` samples exactly that regime and finds no false certificate at $d = 2, 4, 8$. *That evidence should be read for what it is*: across 6,500 trials the sweep drew no contextual family at all at any of the three dimensions, so it establishes the absence of false certificates only vacuously; the positive instances are the two pinned controls. It is a consistency check on the implementation, not a search that could have failed. Note also that restricting to closed triples is the *conservative* move — it is safe whether or not the general-family incompleteness of [2, Prop. 9] is ever witnessed by an explicit example, and we have not verified that it is. Readers should not extrapolate the theorem past this hypothesis.

Both directions of Theorem 1 are now proved: the criterion direction is [2, Thm. 8], and the converse is established by the Smith-normal-form double-annihilator argument recorded in the companion note [6] (verification V44–V46). The 2,100-family stress test in `holonomy_vs_solvability.py` — over 1,500 random families at $d = 2$ and 600 at $d = 4$, unsolvability of $A\gamma \equiv s$ (decided by exhaustive dual search) coinciding with the odd-Q cycle test in every instance, with the Peres–Mermin square ($d = 2$) and the certified $d = 4$ family of [2] pinned as positives so that both truth values of the equivalence are exercised at both d — now stands as an independent numerical cross-check of the analytic result, no longer its sole support. *The random draw should be read honestly*: at $d = 4$ it produced no unsolvable family at all, so the "only if" direction is exercised there by the pinned certificate rather than by the sample. The achievable obstruction values themselves are classified as $\{0, d/2\}$ for even d and $\{0\}$ for odd d; **that classification is due to Abramsky, Cercelescu and Constantin** [4] (upper bound and odd-d vanishing recast in [2, Thm. 1]), and **attainment at general even d is the ROZF construction** [3] (recast in [2, Thm. 7]). Both are used here, not rederived.

3 Both containments are strict

Proposition 1 (Nontrivial class, zero AvN)**.** *On $\mathbb{Z}_3^2$ the Heisenberg cocycle $\omega(u,v) = \zeta_3^{\langle u,v\rangle}$ and its square are coboundaries for none of the $3^9 = 19{,}683$ cochains; the class has order exactly 3. Simultaneously, every cycle of every sampled qutrit Weyl family has even self-pairing, and the achievable AvN value at $d = 3$ is 0 — the odd-d half of the Abramsky–Cercelescu–Constantin classification [4].*

Verified exhaustively in `d3_gap_certificate.py`. Hence a nontrivial canonical class does not imply operational (AvN) contextuality. Proposition 1 is by itself the reason any such program must weaken "encodes" to "detects"; the weakening is forced by this certificate, not adopted out of caution.

Proposition 2 (Contextuality, zero class shadow)**.** *The KCBS pentagon in $d = 3$ attains quantum value $\sqrt{5} \approx 2.2361$ against the exhaustive classical bound 2, while the state-independent class shadow at $d = 3$ is 0.*

Verified in `kcbs_converse.py`. Hence state-dependent contextuality can be invisible to the class: the converse direction of the naive slogan also fails.

4 The Wigner–friend loop is level-split

Consider the friend-qubit loop $T \to T_x \to T_y \to T$ generated by $U_1 = \mathbb{1} \otimes H$, $U_2 = \mathbb{1} \otimes S$.

Observation 1 (Section dependence)**.** *The projective (Pancharatnam) holonomy of the loop is $+1$: the two eigenvector strands carry Bargmann phases $e^{\pm i\pi/4}$ (the Bloch-octant $-\Omega/2$ law), whose product is 1, gauge-invariantly. The determinant-section holonomy is exactly $-i$ for every special-unitary closing arrow; $-i = \tau^{-1}$ at $d = 2$, a τ-level invariant of the 2-adic tower, with section-independent square -1.*

Verified in `wf_loop_holonomy.py` (200 gauge randomizations; 500 closers). Any physical or interpretational claim about "the" holonomy of this loop must state the level; the two answers differ.

Hardware predictions. The strand phases $\pm 45^\circ$ and loop product $+1$ are interferometrically measurable with two-qubit Hadamard tests; on the same device the Peres–Mermin loop yields context signs $(+,+,+,+,+,-)$, i.e. ray-level holonomy -1. One machine, two loops, opposite holonomies, both sharp.

Measured (`ibm_fez`, 2026-07-09, job `d986q62f47jc73a7hm2g`; registry record `hardware/results/ibm_fez_holonomy_20260709.json`): strand phases $+45.88^\circ$ and -48.81° against targets $\pm 45^\circ$ (per-strand statistical error $\approx 1.1^\circ$; the $\sim 3.5\sigma$ offset on the X-prepared strand is a plausible preparation systematic), loop phase $-2.93^\circ \pm 1.55^\circ$ against target 0°. The ray-level holonomy of the Wigner–friend loop is thus measured to be $+1$; the value $-i$ (loop -90°) is excluded at this level by $\approx 56\sigma$, in accordance with Observation 1: $-i$ is a section-level, not interferometric, invariant. On the same device the Peres–Mermin loop gave context products $(+0.750, +0.835, +0.722, +0.780, +0.824, -0.700)$, sign pattern 6/6 as predicted, and $S = 4.6125 \pm 0.0173$: a 35.4σ violation of the exact deterministic noncontextual bound 4 (the run-time script carried the loose $n_c - 1 = 5$ bound, since replaced in `qkernel` by the exhaustive tight bound), consistent with the archived `ibm_fez` value 4.558 ± 0.018.

Hardware caveat. Both significances quoted above are *shot-noise statistical only*, computed under the implemented witness model and conditional on the usual compatibility and nondisturbance assumptions. They are not systematics-aware, and neither measurement is loophole-free; the $\approx 3.5\sigma$ preparation offset on the X-prepared strand is itself evidence that unmodelled systematics are present at a level the statistical error does not capture. We record one further discrepancy rather than suppress it: the pinned value $S = 4.6125$ cannot be recomputed from the stored counts, a naive recomputation giving 4.5662, a gap of about 2.7 standard errors. The archived value 4.558 ± 0.018 is the one an independent reader can reproduce from the deposited data. Read these numbers as consistency checks of the sign predictions on real hardware, not as precision measurements or as certifications.

5 The residual problem

Open Problem 1. *Characterize, in invariants of the context stack $[X/\mathrm{PU}(n)]$ refined beyond the canonical H^2 class, the property "some state has positive contextual fraction on F". Equivalently: identify the cohomological home of KCBS-type (state-dependent, class-invisible) contextuality, and prove or refute that a sheaf-theoretic refinement restores a two-directional equivalence.*

Propositions 1–2 show no formulation at the level of the canonical class alone can succeed; Theorem 1 shows the AvN sector is already optimally captured there.

Lemma 1 (No fully covariant qubit-pair nets)**.** *There is no two-qubit quasi-probability frame, covariant under all Pauli displacements, whose every Lagrangian line-sum is a stabilizer projector: the* 15 *Lagrangian sign conditions have rank* 10 *over* $\mathbb{F}_2$ *and are inconsistent, with minimal certificate a Peres–Mermin-type magic configuration (nine observables, each in exactly two of six contexts,* ε*-product* -1*). The order-*2 *class thus obstructs the global frame itself (`ghw_net_necessity.py`).*

Genuine Gibbons–Hoffman–Wootters nets evade Lemma 1 by retreating to GF(4) translation covariance; sweeping all 1024 of them (`gf4_net_necessity.py`) eliminates global frames in both remaining directions: a state with contextual fraction 0.3462 (the seeded pool's `rand16`, pinned on the `(a')` line of that script's output) on the trivial-class family admits a nonnegative representation in some net (necessity fails), while $T \otimes T$ is negative in every net (best case $-1/4$) with contextual fraction 0 (sufficiency fails); all 60 stabilizer states are nonnegative in some net (construction sanity). Together with the 2^{15}-frame sweep (`net_robust_negativity.py`), every global-frame formulation of the even-d classifier is refuted by explicit certificate; the scenario-paired formulation below is not merely the surviving candidate but the forced one.

The constructive half now exists (`paired_frame_construction.py`). For a closed-triple Weyl family F at $d = 2$, the quantum model $e(\rho)$ is a bijection of the characteristic vector $c(\rho) = (\operatorname{tr} \rho T_v)_{v \in O(F)}$, and every noncontextual model is supported on the solution set $L(F)$ of the family's sign system (parity-violating outcomes have $e \equiv 0$). Hence:

Theorem 2 (Codeword-polytope classification; machine-validated)**.** *For closed-triple Weyl families at $d = 2$, $\mathrm{CF}(F, \rho) = 0 \iff c(\rho) \in \operatorname{conv} L(F)$, the ε-shifted codeword polytope of the family's $\mathbb{F}_2$ kernel; and $L(F) = \varnothing$ exactly on the AvN sector, so one statement covers both sectors. On the two trivial-class Peres–Mermin subfamilies the polytope has exactly* 24 *facets, all triangles $\pm c_u \pm c_v \pm c_w \leq 1$; the classification holds* 40/40 *on both, while the canonical visible frame alone (*15/40*,* 20/40*) and the induced CHSH squares (*31/40*,* 29/40*) are each insufficient — negative results retained as certificates.*

The pairing of Conjecture 1 is thus, in the even-d closed-triple case, membership in a polytope cut out by the same $\mathbb{F}_2$ kernel that classifies the AvN sector; what remains open is general d, mixed families, and the written proof of the facet structure.

The ghost facet theorem and a closed formula

The facet structure now has a proof and yields an exact formula (`ghost_facet_theorem.py`). Let F be Peres–Mermin minus one context C^* with sign ε^*. *Bijection lemma:* for a closed triple with $T_uT_v = \varepsilon_C T_w$, the joint distribution is supported on $s_u s_v s_w = \varepsilon_C$ where $e_C(s) = \frac{1}{4}(1 + \sum_i s_i c_i)$; hence $e(\rho) \leftrightarrow c(\rho)$ affinely. *Support lemma:* noncontextual models put no mass on parity-violating outcomes, so they are mixtures over $L(F)$, and a mixture's model depends only on its mean, giving Theorem 2. *Two-tetrahedra lemma:* every $\lambda \in L$ has ghost parity $\prod_{v\in C^*} \lambda_v = -\varepsilon^*$ (the product of the five remaining signs, $= -\varepsilon^*$ because the six Peres–Mermin signs multiply to -1 — the class), while every quantum state's ghost coordinates lie in the opposite tetrahedron of parity ε^* (since $T_1T_2T_3 = \varepsilon^*\mathbb{1}$ on C^*). Consequently, for σ with $\prod \sigma_i = \varepsilon^*$, every $\lambda \in L$ disagrees with σ in an odd number of positions, so $\sum_i \sigma_i \lambda_{v_i} \leq 1$: these four inequalities are valid, tight on 12 of the 16 vertices, and are the only quantum-operative facets; the 20 in-context facets are the probability positivity conditions $e_C(s) \geq 0$, never violated by quantum states, and the opposite-parity ghost directions are unconstrained (L reaches 3). The classification therefore collapses to a formula:

$$\mathrm{CF}(F,\rho) \;=\; \max\Big(0,\; \tfrac{S^*(\rho)-1}{2}\Big), \qquad S^*(\rho) = \max_{\prod \sigma = \varepsilon^*} \sum_{v\in C^*} \sigma_v \operatorname{tr}(\rho T_v),$$

verified to 10^{-16} across 60 states on each of three deletions. The upper bound $\mathrm{CF} \leq (S^*-1)/2$ follows analytically from the validity above; equality now follows from the *Completion Lemma*: for each operative pattern σ the point y_σ with ghost coordinates σ and all others zero is a valid parity-supported no-signalling model (every remaining context contains at most one ghost observable, so its probabilities are $(1 \pm 1)/4$ or $1/4$), attaining $\sigma \cdot c = 3$; Abramsky–Barbosa–Mansfield duality then gives $\mathrm{CF} = \max_\sigma (S_\sigma - 1)/(3-1)$ exactly. *What this proves, exactly:* the closed-form classification is established end to end *for the deletion families treated here* — Peres–Mermin with one context removed — upper bound analytically, equality by the Completion Lemma. Theorem 2 as stated quantifies over all closed-triple Weyl families at $d = 2$, and in that generality it is *machine-validated* (40/40 on both subfamilies), not proved; the general compressed-coordinate statement is recorded as open in the verification ledger (V52). We label the theorem accordingly and do not close that gap here. Corollary: joint eigenstates of C^* with an operative pattern attain $S^* = 3$ and hence $\mathrm{CF} = 1$ — for the deleted odd column these are the Bell states, so *maximal contextuality survives the deletion of the very context that carried the class*: the class does not disappear, it flips the ghost tetrahedron.

The ghost at $d = 4$: the tower appears

The mechanism lifts to the first nontrivial tower level (`d4_ghost_tower.py`), on the certified $d = 4$ magic square (six contexts, nine observables of mixed orders 2 and 4, each observable in exactly two contexts, phase exponents summing to $d/2$ — the class value). Deleting any context C^* leaves a solvable system with $|L| = 64$ spectral assignments, and the ghost value $\sum_{v\in C^*} \lambda_v$ is constant on L with

$$\gamma - s^* \;\equiv\; d/2 \pmod{d}:$$

the classical ghost torus is rotated from the quantum one by exactly the top 2-adic layer $\omega^{d/2} = -1$, for both a pure order-2 ghost and a mixed order-2/4 ghost — and the $d = 2$ opposite-tetrahedra lemma is the same law, since there too the offset was $d/2$. Every joint eigenstate class of the deleted context attains CF $= 1.0000$ exactly (4 and 8 classes respectively), and Haar-random states populate the sector (sampled CF up to 0.21). Two negative results are pinned honestly: noncontextual assignments must be restricted to spectral values (non-spectral assignments hit zero-probability outcomes; the unrestricted relaxation under-computes CF), and the $d = 4$ law is *not* a function of the Hermitian order-2 shadows of the ghost alone — the Hermitianized $d = 2$ formula predicts zero precisely where random states reach CF $= 0.138$. The exact $d = 4$ classification is polytope membership in the full moment space of Theorem 2. Two further results pin its structure (`d4_moment_facets.py`).

Ghost-law lemma (now proved). Since every observable lies in exactly two contexts, summing the remaining constraints over any $\lambda \in L$ gives $\gamma = \sum_{\text{rem}} s - 2T \pmod d$, with $T = \sum_{v \notin C^*} \lambda_v \bmod 2$; using $\sum_{\text{all}} s = d/2$ (the class) this reads $\gamma - s^* = d/2 - 2(s^* + T)$. T is constant on L and $s^* + T$ is even for every one of the six deletions (machine-verified), so $\gamma - s^* = d/2$ always: the tower ghost law is an identity plus a verified parity sub-lemma.

τ-necessity (twelve explicit certificates). For each sampled state with CF > 0, the *optimal* separating functional restricted to the even-character subspace — all order-2 shadow data, i.e. everything visible below the top 2-adic layer — achieves gap ≤ 0: the even projection of the quantum moment vector lies *inside* the even projection of the classical polytope. Hence the $d = 4$ state sector is invisible at the order-2 layer, and the τ-level moments are necessary for every operative facet. The 2-adic tower is not bookkeeping; it is load-bearing for the state sector, exactly as the even-d half of Conjecture 1 asserts. The odd-sector facet structure is now partially decoded (`d4_odd_sector_facets.py`). The polytope has affine dimension 33; sparse certificate mining shows the operative geometry is two-tiered. *Tier one* consists of triangle inequalities on *closed τ-twisted order-2 triples*: every 3-sparse certificate coordinate (21/21) is an order-2 operator reached through odd character indices of its host context — the shadow wearing the top-layer cocycle phase — and the recurring triples are the deleted context C^* itself (the τ-twisted ghost) and a virtual closed triple $\{(0,2,2,0),(2,0,0,2),(2,2,2,2)\}$. This tier classifies 32/40 sampled states. *Tier two*, required by the remaining contextual states, involves genuinely order-4 operator moments — the second level of the tower — and is now pinned exactly (`d4_tier2_catalogue.py`, corrected by `d4_arithmetic_profile.py`): the extracted supporting inequalities have tight-vertex rank exactly 32 (true facets of the 33-dimensional hull), and each admits an exact $\{0, \pm 1\}$-coefficient lift attaining an *integer* bound, 6 or 7, with equality on its tight set. The tier-two data is rational throughout.

Retraction 1. An earlier draft presented these facets on a $\frac{1}{2\sqrt{2}}\mathbb{Z}$ grid with a single coefficient magnitude $3/(2\sqrt{2})$, and read the resulting unit-coefficient bounds $4\sqrt{2}$ and $14\sqrt{2}/3$ as placing Tsirelson's $\sqrt{2}$ inside the tier-two geometry. The whole presentation was a normalization artifact and is withdrawn: not only the reading, but the grid and the coefficient magnitude. The stored lift has coefficients in $\{0, \pm 1\}$, which do not lie on that grid at all, and no vector of constant magnitude $3/(2\sqrt{2})$ can attain a nonzero integer bound on integer vertices. The artifact arose from a rounding collision in the earlier magnitude bookkeeping, where $|\pm 1| \cdot 2\sqrt{2}$ and $|3/(2\sqrt{2})| \cdot 2\sqrt{2}$ both round to 3; the $\sqrt{2}$ was injected by that convention and by nothing in the geometry. Consequently every facet functional in the census is rational, and $\mathbb{Q}(\sqrt{2})$ is not among the eighteen canonical arithmetic fields of the 61 classes: it survives only in the auxiliary 33-coordinate metric on the bound-7 classes $\{48, 49, 52\}$ (which are bound-7 classes, though not the only ones). Certificate: `d4_arithmetic_profile.py`.

Together, tier one and the exact tier-two facets classify 40/40 states of the construction sample; a subsequent fresh-sample audit shows this is *not yet the complete family* — new weak-band contextual states require further tier-two orbit members. Closing the extracted seeds under the two proven vertex symmetries (kernel translations and global conjugation, ~ 1152 facets) recovers most but not all fresh states, and per-context outcome relabelings are provably *not* symmetries (each observable lives in two contexts). The deleted family's automorphism group is now derived: the family is $K_{3,3}$ (contexts as nodes, observables as edges), and the automorphisms fixing the deleted context form the twelve graph automorphisms dressed by a 64-torsor of spectral shifts, times conjugation — order 1536, each element provably a bijection of the 64 vertices, acting on moment space by exact monomial rotations. The census is now *complete and exact.* The polytope — 64 integer $\{-1, 0, 1\}$ points in faithful 33-dimensional integer coordinates — has exactly **23,256** facets in **61** symmetry classes, derived twice by independent methods that agree class-for-class: an adjacency decomposition (breadth-first closure of the facet–ridge graph modulo the derived group, every pivot certified by exact integer nullspace recovery through rank-32 tight sets and exact validity on all vertices), and a direct exact enumeration of the full polytope by Normaliz. Tight-vertex counts run 36–60; orbit sizes run 4 to 768, so the census contains both highly symmetric facets (stabilizers of order 192) and free ones. Since quantum moment vectors lie in the affine hull (residual $\sim 10^{-14}$), the catalogue is the classification theorem in operational form: $\mathrm{CF}(\rho) > 0$ if and only if $\mu(\rho)$ violates one of the 23,256 inequalities; on fresh sampled states the facet classifier agrees with the LP 40/40, as it must. Two method notes are pinned in the suite: `lrs` wedges reproducibly on several of these tight-set polytopes (stalling at identical partial output under different budgets) while `cdd` and Normaliz succeed; and an intermediate in-walk count of "121 classes" was a bookkeeping duplication (two key encodings double-storing each orbit) exposed only by the independent cross-check — the cross-check is not optional. The tower appears twice over: as the ghost rotation $d/2$, and as the stratification of operative facets by operator order. (An earlier draft counted a third appearance, a $\mathbb{Z}[1/\sqrt{2}]$ arithmetic of the tier-two facet data; that reading is withdrawn with Retraction 1 — the tier-two data is rational.) The arithmetic profile of all 61 classes is now complete: exactly 18 arithmetic species, classified by the pair (integer bound, squared norm of the exact minimal-norm lift), all functionals rational. Open: the no-signalling-normalized closed formula, and mixed families.

Conjecture 1 (State-sector pairing)**.** *For odd prime d, the state-dependent sector is classified by the pairing of the state against the order-d class through the Weyl transform: a state yields positive contextual fraction on stabilizer measurement scenarios if and only if its parity-frame quasi-distribution is negative — the Howard–Wallman–Veitch–Emerson contextuality–negativity correspondence [5], recast as a stack pairing. For even d, where no Gross frame exists, the corresponding role is played by the top layer of the 2-adic tower of [2].*

Computed anchors (`state_sector_probe.py`): $\mathrm{CF}(\mathrm{PM}) = 1.0000$ for every state (AvN); $\mathrm{CF}(\mathrm{KCBS}) = 0.4721 = 2\sqrt{5} - 4$ at the optimal state with class shadow 0; at $d = 3$ the parity-frame quasi-distribution is nonnegative on all twelve stabilizer pure states (exhaustive) and attains $-1/3$ on the strange state. Notably, the τ-convention frame of [2] exhibits $-1/18$ pseudo-negativity on stabilizer states at $d = 3$: τ is the even-d object, and the odd/even divide of [1] resurfaces at the level of quasi-probability frames.

A Claims-to-verification table

Claim	Script	Expected one-liner
Thm. 1	`holonomy_vs_solvability.py`	0 mismatches; pinned PM/cert4 agree
Prop. 1	`d3_gap_certificate.py`	order 3; AvN $\{0\}$; gap certified
Prop. 2	`kcbs_converse.py`	$2.23607 > 2$; converse gap certified
Obs. 1	`wf_loop_holonomy.py`	strands $\pm\pi/4$; det-holonomy $-i$
PM anchor	`pm_gauge_invariance.py`	product -1 over all 512 gauges

Note on methods. Large-language-model assistants (Anthropic Claude) were used substantially in drafting this paper, in developing the verification scripts, in literature search and in proposing proofs. Every load-bearing claim is machine-verified independently of that assistance by the scripts named in the text, and the author takes full responsibility for the results, including any errors.

References

[1] M. Flores Gordillo, *The Canonical Class of the MASA Context Stack and the Peres–Mermin Obstruction*, Zenodo, 2026. `doi:10.5281/zenodo.21630431`

[2] M. Flores Gordillo, *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower*, Zenodo, 2026. `doi:10.5281/zenodo.21630630`

[3] R. Raussendorf, C. Okay, M. Zurel, P. Feldmann, *Quantum* **7**, 979 (2023), arXiv:2110.11631.

[4] S. Abramsky, Ş.-I. Cercelescu, C. Constantin, *Commutation Groups and State-Independent Contextuality*, in *9th International Conference on Formal Structures for Computation and Deduction (FSCD 2024)*, LIPIcs **299**; extended version arXiv:2603.12197 (2026).

[5] M. Howard, J. Wallman, V. Veitch, J. Emerson, *Contextuality supplies the magic for quantum computation*, Nature 510, 351 (2014).

[6] M. Flores Gordillo, *Local validity and contextual holonomy: a structural formulation of the Copenhagen principle*, companion note v3, Zenodo, 2026. `doi:10.5281/zenodo.21630739`

[7] `github.com/manuflog/contextuality-obstructions`, verification suite, 2026.